

EDUCATION

- Sep. 2021 – Present Doctor of Philosophy in Computer Science, **Hong Kong Polytechnic University**
Supervisor: Dr. Jieming Shi **Co-Supervisor:** Prof. Man Lung Yiu
Thesis: Trajectory Data Management: From Sparse Points to Smart Mobility
- Sep. 2017 – Jul. 2020 Master of Engineering in Computer Technology, **Sichuan University**
Supervisor: Prof. Min Zhu
Thesis: Research on Task-driven Visualization Recommendation for Relational Data
- Sep. 2013 – Jul. 2017 Bachelor of Engineering in Computer Science and Technology, **Sichuan University**
Avg Score: 87.52 / 100 **GPA:** 3.52 / 4.0 **Rank:** 24th out of 361 **First Class Honors**

WORKING EXPERIENCE

- 2020.08 – 2021.06: **Research Intern**, Database and Storage Lab, Alibaba DAMO Academy.
- 2019.08 – 2019.12: **Visiting Student**, Big Data and Database Group, RMIT University.
- 2018.07 – 2018.12: **Internship**, Department of Social and Performance Ads, Tencent.
- 2016.07 – 2016.10: **Internship**, Department of Social Network Operation, Tencent.

RESEARCH INTERESTS

Spatial and Temporal Data Management, Urban Computing, Geo-spatial Data Analysis and Visualization

PUBLICATIONS

- **Wei Tian**, Jieming Shi, Man Lung Yiu. [UNITE: A Unified Framework for Accurate and Efficient Origin-Destination and Route Travel Time Estimation](#). **SIGKDD 2026**, to appear. (CCF-A, CORE A*)
- **Wei Tian**, Jieming Shi, Man Lung Yiu. [Efficient Methods for Accurate Sparse Trajectory Recovery and Map Matching](#). *IEEE International Conference on Data Engineering (ICDE)*, 2025, pp. 363–375. (CCF-A, CORE A*)
- **Wei Tian**, Jieming Shi, Siqiang Luo, Hui Li, Xike Xie, Yuanhang Zou. [Effective and Efficient Route Planning Using Historical Trajectories on Road Networks](#). **PVLDB 2023**, 16(10): 2512–2524, 2023. (CCF-A, CORE A*)
- Min Zhu, **Wei Tian**, Di Peng, Yabo Su, Hao Niu. [Method of Predicting User Professionalism Based on Question Granularity in Community Question Answering](#). *Advanced Engineering Sciences*, 2019, 51(1): 173–180. (EI)
- Hai Lan, Jiong Xie, Zhifeng Bao, Feifei Li, **Wei Tian**, Fang Wang, Sheng Wang, Ailin Zhang. [VRE: A Versatile, Robust, and Economical Trajectory Data System](#). **PVLDB 2022**, 15(12): 3398–3410 (2022). (CCF-A, CORE A*)
- Di Peng, **Wei Tian**, Min Zhu, Yukun Ren, Xiaojian Lin, Mingzhao Li. [TargetingVis: Visual Exploration and Analysis of Targeted Advertising Data](#). *Journal of Visualization*, 2020, 23: 1113–1127. (SCI)
- Di Peng, **Wei Tian**, Binbin Lu, Min Zhu. [DMNEVis: A Novel Visual Approach to Explore Evolution of Dynamic Multivariate Network](#). *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2018, pp. 4304–4311. (CCF-C, CORE B)
- Min Zhu, Jianhua Wang, Xiaowei Li, **Wei Tian**. [Fast Percentage Closer Soft Shadows](#). *Advanced Engineering Sciences*, 2019, 51(4): 140–146. (EI)
- Min Zhu, Li Chen, Qin Li, **Wei Tian**, et.al. [Book: Handbook of Computer Network Course Projects \(in Chinese: 计算机网络课程设计\)](#). *China Machine Press (机械工业出版社)*, ISBN: 978-7-111-61139-4, 2018.

PATENTS

- 朱敏, 李龙兴, **田伟**. 一种基于联合嵌入的任务驱动关系型数据视图推荐方法: 中国, 202111527276.1[P].
- 朱敏, 彭第, 韦东鑫, **田伟**, 李亚男, 任宇堃, 林晓健. 层次数据的关联关系可视化处理方法、服务器及存储介质: 中国, 201911287989.8[P].
- 朱敏, 李亚男, 彭第, 游思兰, 韦东鑫, **田伟**. 一种气泡层次可视化中气泡分离方法: 中国, 201811300412.1[P].

TEACHING EXPERIENCE

- 2024 Spring: **Tutor**, COMP1011 - Programming Fundamentals, Hong Kong Polytechnic University
- 2023 Fall: **Teaching Assistant**, COMP5434 - Big Data Computing, Hong Kong Polytechnic University
- 2023 Spring: **Teaching Assistant**, COMP4434 - Big Data Analytics, Hong Kong Polytechnic University
- 2022 Fall: **Teaching Assistant**, COMP5434 - Big Data Computing, Hong Kong Polytechnic University

- 2022 Spring: **Teaching Assistant**, COMP5434 - Big Data Computing, Hong Kong Polytechnic University
- 2021 Fall: **Teaching Assistant**, ENG2002 - Computer Programming, Hong Kong Polytechnic University
- 2019 Summer Term: **Tutor**, Urban Computing, Sichuan University
- 2018 Summer Term: **Tutor**, Big Data Management, Sichuan University
- 2017 Fall: **Teaching Assistant**, Course of Computer Network, Sichuan University

RESEARCH FUNDINGS

- **Participator**, Large-Scale Multilayer Graph Management: Queries, Indexes, and Algorithms. *Hong Kong General Research Fund*, 2025.1-2027.12, HKD 878,772.
- **Participator**, 面向大规模复杂图数据的异常检测技术研究. *NSFC Young Scientists Fund*, 2023.1-2025.12, CNY 300,000.
- **Participator**, GPU-based Approximate Query Processing of High-Order Proximities on Billion-Scale Graphs. *Hong Kong RGC Early Career Scheme*, 2022.1-2025.6, HKD 740,067.
- **Participator**, A Distributed Elastic Data Processing System for Massive Spatio-temporal Trajectories. *Alibaba Innovative Research (AIR) programme*, 2021.1-2021.12, CNY 200,000.
- **Participator**, Research on Selection Pattern of Advertising Targets Based on Visual Analysis. *Tencent Rhino-bird Program*, 2018.5-2019.5, CNY 200,000.
- **Participator**, Visual Recommendation System Based on User-generated Content of Tourism. *Sichuan Science and Technology Program of China*, 2018.1-2019.12, CNY 200,000.

REPRESENTATIVE AWARDS

- The Hong Kong Polytechnic University Research Postgraduate Scholarship (2021)
- Outstanding Master Graduates of Sichuan University (2020)
- First-class of Excellent Postgraduate Scholarship by Sichuan University (2019)
- Outstanding Postgraduate of Sichuan University (2019)
- HUAWEI Scholarship of Sichuan University (2018)
- Outstanding student member of Sichuan University Branch of CCF (2018)
- Honorable mention of ChinaVis Data Challenge (Qingdao'2017)
- Outstanding Student of Sichuan University (2016)
- 2×National Encouragement Scholarship (2014, 2016)
- 2×Third-class of Excellent Undergraduate Scholarship by Sichuan University (9%) (2014, 2016)
- Second Prize of Sichuan Regional Contest of 7th Lan Qiao Cup (2016)

ACADEMIC ACTIVITIES

- External Reviewer for
 - IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC'2018)
 - China Visualization and Visual Analytics Conference (Chengdu'2019, Qingdao'2017)
- Volunteer of Academic Conference
 - IEEE International Conference on Data Engineering (Hong Kong'2025)
 - China Visualization and Visual Analytics Conference (Chengdu'2019)
 - Big Data Visualization Summit Forum (Chengdu'2017)
- Invited Talks
 - Jun. 2021: Vision Computing Lab, Sichuan University, China
- Membership
 - CCF (China Computer Federation) Student Member
 - ACM (Association for Computing Machinery) Student Member

RESEARCH PROJECTS

- Aug. 2020 – Jun. 2021* *NoSQL Storage Engine for Big Trajectory Data Management and Analysis*
Research Intern in Database and Storage Laboratory, **Alibaba DAMO Academy**.
- ❖ Propose a Segment-based model to storing massive trajectory data, which saves about 3x – 4x storage cost.
 - ❖ Based on the Segment-based model, make the query performance of five basic categories 2x faster than existing methods.
 - ❖ Based on HBase, develop a storage engine for trajectory data.

- Aug. 2019 – Dec. 2019* *Research on visualization recommendation for relational data based on visual analytical task*
ISAR (Information Storage, Analysis and Retrieval) Research Group, **RMIT University**.
Visiting Student **RMIT Supervisor:** Associate Professor Zhifeng Bao
- ❖ Propose an end-to-end joint embedding learning framework, which aims to learn the association between Data-Task and visualization Views.
 - ❖ Propose a recommendation algorithm for visualization by fusing dataset features, analytical task features and visualization features with attention mechanism.
- Jan. 2019 – Apr. 2019* *A Carpooling Algorithm Based on Region Division Using Trajectory Data*
- ❖ Propose a dividing urban area algorithm based on trajectory data by considering traffic conditions and semantic information in sub-regions.
 - ❖ Propose a new carpooling model based on sub-regions by pruning the search space of matching riders and vehicles in region-level.
- May 2018 – May 2019* *Research on selection pattern of advertising targets based on visual analysis*
Project supported by **Tencent Rhino Bird Program** (Grant No. 18H0487) (¥200,000)
Research Assistant
- ❖ Model the advertising behaviors of advertisers with multiple dimensions, such as industry type and product type.
 - ❖ Analyze effect and relationship of targets, discovering classic cases and patterns in targets system.
- Jan. 2018 – Dec. 2019* *Visual recommendation system based on user-generated content of tourism*
Project supported by **Sichuan Science and Technology Program of China** (Grant No. 2018GZ0171) (¥200,000)
Research Assistant
- ❖ Mine the frequent patterns of travel sequences from the massive travel blog contents by using NLP methods.
 - ❖ Detect tourism area and popular local trip route.
- Feb. 2018 – May 2018* *Visual Approach to Explore Evolution of Dynamic Multivariate Network*
- ❖ Propose a network visualization method based on community partitioning and hierarchical progress (group network exploration).
 - ❖ Design a vertical stacking and adjustable network visualization method (ego-centric network exploration)
- Oct. 2017 – Jan. 2018* *Expert Recommendation in Community Question Answering*
- ❖ Classify expertise metrics into community-level, topic-level and question-level, then propose a question-level expertise modeling method.
 - ❖ Propose an expert recommendation algorithm by incorporating authority, interest and expertise.

REFERENCE

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- **Prof. Jieming Shi**, Email: jieming.shi@polyu.edu.hk
 - **Prof. Man Lung Yiu**, Email: csmlyiu@comp.polyu.edu.hk
 - **Prof. Min Zhu**, Email: zhumin@scu.edu.cn
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